Activity 3: Buried Sphere Width–Depth

Given:

$$g_z(x) = \frac{4\pi G}{3} \Delta\rho\, a^3 \frac{z_c}{(x^2 + z_c^2)^{3/2}}.$$

Tasks

1. Show that $g_z(0)$ is proportional to $\Delta\rho\, a^3 / z_c^2$.
2. Solve for $x_{1/2}$ such that $g_z(x_{1/2}) = \frac{1}{2} g_z(0)$.
3. Derive:

   $$z_c \approx 0.65\, w$$

   where w is the full-width half-maximum. Hint: this is on the right track,

   $$\frac{z_c}{x_{1/2}} = (2^{2/3} - 1)^{-1/2}.$$

4. Explain:
   - Why depth affects width but density contrast does not.
   - How the radius and density contrast result create non-uniqueness for inversion and how we ensure uniqueness.
   - Why a shallow lenticular body could mimic a deep sphere's anomaly.